



POLITECNICO

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Financial Engineering

Assignment 4

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Individual Solution

Contents

1	Case study: Certificate Pricing	1
a	Methodology	1
b	Results and Discussion	1
2	Exercise 2: Pricing a Digital Option	2
a	Methodology	2
b	Results and Discussion	2
3	Exercise 3: Pricing under a Given Characteristic Function	3
a	Pricing Methodologies	3
b	Results and Discussion	3
4	Exercise 4: Option Pricing under NTS Exponential Lévy Models	4
a	Methodology	4
b	Results and Discussion	5
5	Case study: Volatility Surface Calibration	5
a	Methodology	5
b	Results and Discussion	6

1. Case study: Certificate Pricing

a. Methodology

This section prices an Equity Protection Certificate issued by Bank XX on February 15th, 2008, with a principal of €100 million, a 5-year maturity, a 95% protection level, and an exotic coupon paying a 110% participation on the positive performance of an equally weighted arithmetic basket of ENI and AXA. The certificate is valued under a single-curve framework, assuming zero counterparty risk (ISDA agreement with CSA).

Structuring and swap valuation.

The issuer (Party A) hedges the exotic leg by entering a swap with the Investment Bank (Party B). Party A pays the Euribor 3m + spread floating leg and a maturity payment of $(1 - P)N$; Party B pays the upfront $X\%$ at inception and the exotic coupon at maturity. Equating the two legs' NPV at t_0 gives

$$X\% = 1 - B(t_0, t_N) + s \cdot \text{BPV} + (1 - P)B(t_0, t_N) - c, \quad (1)$$

where $c = C/N$ is the normalized option value, $B(t_0, t_N)$ is the discount factor at maturity, and BPV is the Act/360 basis point value of the floating leg computed over the adjusted payment schedule.

Exotic coupon pricing.

The coupon is a European call on an equally weighted arithmetic basket, with payoff $N \cdot \alpha \cdot \max(\text{Basket}_{t_N} - P, 0)$. Since a sum of lognormal variables is not itself lognormal, no closed form exists; the coupon is priced with two independent methods.

Moment matching. The arithmetic basket is approximated by a synthetic lognormal distribution matched on its first two raw moments M_1, M_2 . From $M_2/M_1^2 = \exp(\Sigma^2 T)$ one recovers the synthetic volatility $\Sigma^2 = \frac{1}{T} \ln(M_2/M_1^2)$, and the coupon is priced with the standard Black-76 formula using M_1 as forward and P as strike.

Monte Carlo. As a cross-check, the correlated ENI/AXA paths are simulated under the risk-neutral measure via the exact GBM solution, with the correlation ρ injected through a Cholesky-transformed pair of standard normals. The basket payoff is averaged and discounted over $N_{sim} = 10^7$ paths.

Both the closed-form upfront $X\%$ and its Monte Carlo counterpart are reported side by side, together with the Monte Carlo standard error, as an explicit validation of the moment-matching approximation.

b. Results and Discussion

Methodology	Coupon Value (EUR)	Final Upfront ($X\%$)
Moment Matching (Analytical)	18,120,357.15	9.9586%
Monte Carlo Simulation	18,110,857.15*	9.9682%
Absolute Difference	9,500.00	0.0095%

Table 1: Structured coupon value and resulting upfront fee: Moment Matching vs. Monte Carlo. *Not printed directly; derived from the two printed upfront percentages via Eq. (1).

The Moment Matching approximation and the Monte Carlo cross-check agree to within €9,500 on an €18.1M coupon (0.05% relative), and to within 1 bp on the resulting upfront fee (9.9586% vs. 9.9682%), confirming that the lognormal moment-matching approximation of the arithmetic basket is highly accurate for this certificate. The Monte Carlo standard error on the coupon estimate is EUR 8,991.26 ($N_{sim} = 10^7$), of the same order of magnitude as the observed discrepancy between the two

methods, confirming that the 9,500€ gap is consistent with simulation noise rather than a structural mismatch between the true basket distribution and its lognormal approximation.

2. Exercise 2: Pricing a Digital Option

a. Methodology

A cash-or-nothing digital call on the Eurostoxx 50 is considered (notional €10 million, payoff 5% of notional, 1-year Act/365 expiry, curve as of February 15th, 2008). The goal is to quantify the pricing gap between a flat-volatility Black (1976) price and a smile-consistent price.

Under Black, the digital price is the negative strike-derivative of the vanilla call, $dc_B(K) = B(t_0, t) \mathcal{N}(d_2)$, with the ATM volatility obtained by cubic-spline interpolation of the market smile at $K = F_0$. Since the flat-volatility assumption ignores the market skew, the digital is reinterpreted as the limit of a bull spread, $dc(K) = -\frac{d}{dK} c[K, \sigma(K)]$, which by the chain rule yields the smile-adjusted price

$$dc_{\text{smile}}(K) = dc_B(K) - \frac{\partial \sigma(K)}{\partial K} \text{Vega}, \quad \text{Vega} = B(t_0, t) F_0 \varphi(d_1) \sqrt{\tau}. \quad (2)$$

The volatility slope at the ATM point is estimated with a central finite difference on the cubic-spline interpolant of the market smile.

b. Results and Discussion

Method	Unit Price	Price (EUR)
Black	0.435235	217,617.60
Smile-Adjusted	0.522392	261,196.25
Difference	0.087157	43,578.65

Table 2: Digital option pricing results: Black vs. smile-adjusted approach.

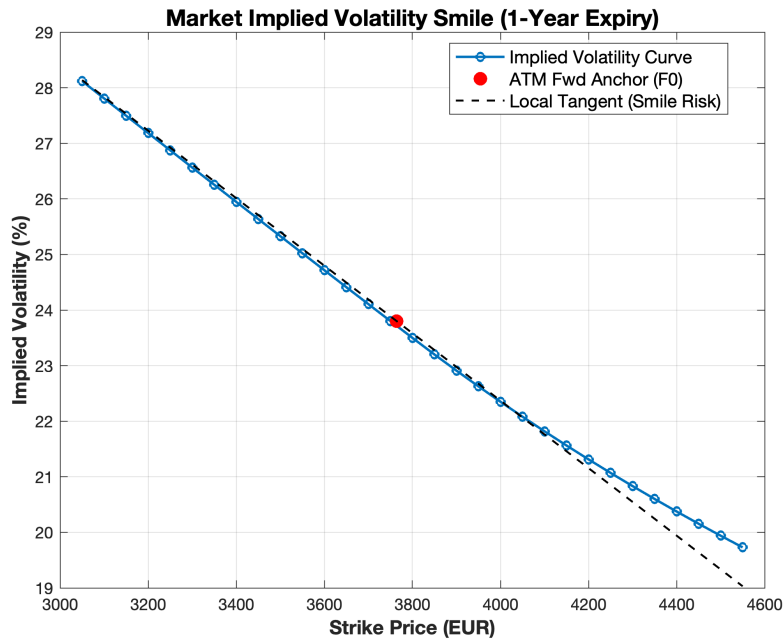


Figure 1: Market implied volatility smile (1-year expiry, Eurostoxx 50), with the ATM forward anchor and the local tangent whose slope drives the smile correction.

At the ATM forward $F_0 = 3763.97$ ($\sigma_{\text{ATM}} = 23.72\%$), the market smile has a negative slope $\partial\sigma/\partial K \approx -6.1 \times 10^{-5}$, consistent with the negative skew typically observed on the Eurostoxx 50. Since the smile correction term is $-(\partial\sigma/\partial K) \cdot \text{Vega}$ with $\text{Vega} = 1435.38$, a negative slope produces a positive correction, pushing the smile price (EUR 261,196.25) above the flat-volatility Black price (EUR 217,617.60) by EUR 43,578.65 ($\sim 20\%$). This confirms that pricing the digital under a flat-volatility assumption systematically underestimates its fair value whenever the underlying exhibits a skewed smile.

3. Exercise 3: Pricing under a Given Characteristic Function

a. Pricing Methodologies

The forward log-return is specified directly through the characteristic function

$$\phi(u) = \frac{1}{\left(1 - i\frac{u}{p^+}\right) \left(1 + i\frac{u}{p^-}\right)} e^{i\mu u}, \quad p^+ = 1.5, \quad p^- = 0.9, \quad (3)$$

with μ fixed by the martingale condition $\phi(-i) = 1$. Four independent pricing routes are implemented and cross-checked on three log-moneyness levels using the Eurostoxx 50 inputs of Exercise 2.

Quadrature.

The Lewis (2001) representation $C(x) = B(t_0, t) F_0 \left[1 - \frac{e^{-x/2}}{2\pi} \int_{-\infty}^{\infty} e^{-i\xi x} \frac{\phi(-\xi - i/2)}{\xi^2 + 1/4} d\xi \right]$ is evaluated point-wise with adaptive Gauss-Kronrod quadrature and used as the numerical benchmark.

FFT.

The same integral is cast on a uniform grid of $N = 2^M$ points via the FFT, using the Nyquist relation $dx \cdot dz = 2\pi/N$. The implementation accepts either the log-moneyness grid spacing x_1 or the frequency-domain spacing dz as a free input and derives the other automatically from the Nyquist relation, rather than fixing both a priori.

Residues.

Since ϕ is rational, the Lewis integral is evaluated in closed form by closing the contour in the complex plane and summing the residues of the poles $z = \pm i/2$ (from the kernel) and $z = i$, $z = -1.4i$ (from ϕ). The contour is closed in the upper half-plane when $x + \mu < 0$ and in the lower half-plane otherwise, consistent with the decay direction of the exponential factor.

Monte Carlo.

Two independent simulation routes are used. *Direct simulation* recognizes ϕ as the characteristic function of $\mu + Y^+ - Y^-$ with $Y^+ \sim \text{Exp}(p^+)$, $Y^- \sim \text{Exp}(p^-)$ (a Double-Exponential/Laplace law), and simulates the terminal log-return exactly with no discretization. *Indirect simulation* reconstructs the CDF of the log-return on a grid from the FFT-priced digitals ($\bar{P} = 1 - \bar{D}$), retains only the region where the reconstructed CDF is monotone, and draws samples by linear-interpolation inversion of uniform draws.

b. Results and Discussion

Using the analytical Residuals solution as reference, both Quadrature and FFT agree to near machine precision, with maximum absolute errors of 1.84×10^{-4} and 5.28×10^{-6} respectively. Direct Monte Carlo shows the expected simulation noise, with a maximum absolute error of 1.09. The two FFT parameterizations (fixed x_1 vs. fixed dz) agree with each other to 5.28×10^{-6} , confirming the correctness

x	K	Quadrature	FFT	Residuals	Direct MC
-5.22%	3965.72	1560.7371	1560.7369	1560.7369	1561.7338
0%	3763.97	1602.0325	1602.0324	1602.0324	1603.0448
+15%	3239.63	1726.8055	1726.8053	1726.8053	1727.8988

Table 3: Double-Exponential call prices (EUR) across all pricing methodologies.

of the automatic Nyquist-relation derivation. The indirect Monte Carlo estimator (inverse-transform sampling from the FFT-reconstructed CDF, Fig. 2–3) deviates from the direct estimator by at most 25.9 under the x_1 -fixed grid and 3.3 under the dz -fixed grid, reflecting a coarser reconstruction of the CDF tail under the former parameterization.

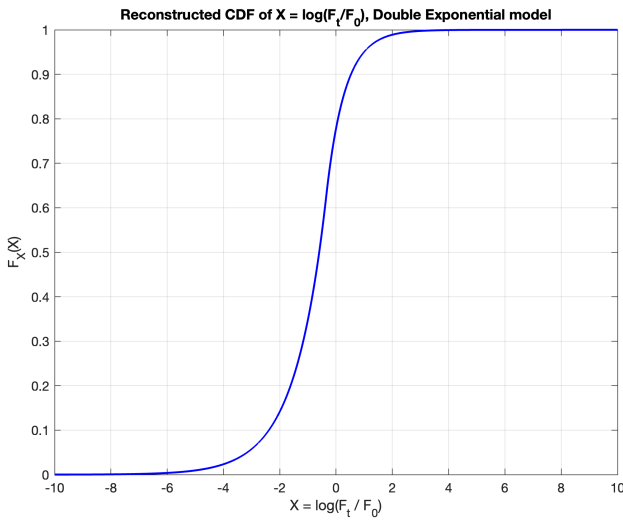


Figure 2: Reconstructed CDF of $X = \log(F_t/F_0)$, Double-Exponential model.

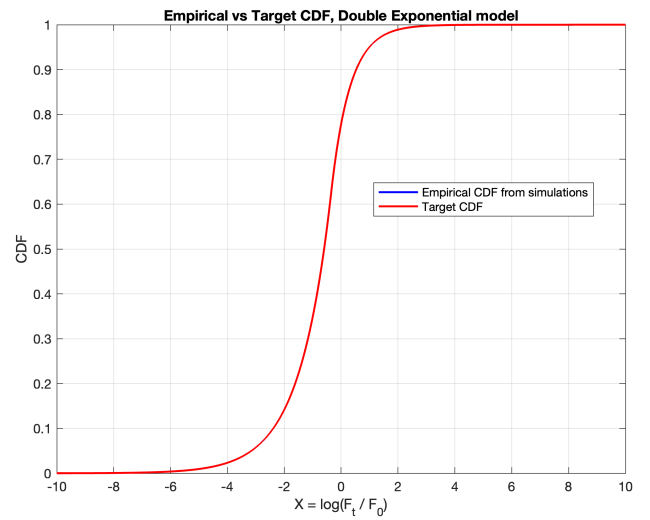


Figure 3: Empirical vs. target CDF, validating the inverse-transform sampling.

4. Exercise 4: Option Pricing under NTS Exponential Lévy Models

a. Methodology

The forward log-return follows a one-step normal mean-variance mixture, $f_t = \sqrt{\Delta t} G \sigma g + \varphi \Delta t G + \mu$, $g \sim \mathcal{N}(0, 1)$, where G is a positive mixing variable with Laplace exponent parametrized by $(\sigma, \kappa, \eta, \alpha)$. The characteristic exponent is implemented as a single function of α ,

$$\psi_\alpha(u) = \frac{\Delta t}{\kappa} \frac{1 - \alpha}{\alpha} \left[1 - \left(1 + \frac{u \kappa \sigma^2}{1 - \alpha} \right)^\alpha \right], \quad (4)$$

so that the Normal Inverse Gaussian case ($\alpha = 1/2$) and the Normal Tempered Stable case used later in Exercise 5 ($\alpha = 2/3$), as well as the facultative $\alpha = 1/3$ case, are all obtained from the same routine rather than three separately hard-coded closed forms.

Prices are computed for $\alpha = 1/2$ (NIG, $\sigma = 20\%$, $\kappa = 1$, $\eta = 3$, $\tau = 1y$) over $x \in [-25\%, 25\%]$ via FFT, quadrature, and direct Monte Carlo sampling of the NIG law using the Michael-Jones-Schucany algorithm on the associated Inverse Gaussian mixing variable ($\mu_{IG} = t$, $\lambda_{IG} = t^2/\kappa$). The IG-moment identities implied by this parametrization are verified by comparing their theoretical values against the empirical mean, variance, skewness, and excess kurtosis of 10^7 Monte Carlo draws of the Inverse Gaussian mixing variable. The facultative $\alpha = 1/3$ case reuses the same pricing routines with the same (σ, κ, η) .

b. Results and Discussion

x	K	Quadrature	FFT	Direct MC
-25%	4833.03	27.8977	27.8977	27.9097
-15%	4373.11	81.2338	81.2338	81.2554
-5%	3956.95	201.8955	201.8955	201.9010
+5%	3580.40	394.3881	394.3881	394.3444
+15%	3239.68	627.6176	627.6176	627.5692
+25%	2931.38	871.9244	871.9244	871.8981

Table 4: NIG call prices (EUR) at selected moneyness values ($x = 0\%$ was not among the sampled diagnostic points printed by `compareComprehensiveMethods`-style reports for this exercise).

FFT and Quadrature agree closely, with a maximum absolute error of 6.60×10^{-4} ; the two FFT parameterizations (fixed x_1 vs. fixed dz) agree with each other to the same order, 6.60×10^{-4} . Direct Monte Carlo shows a maximum absolute error of 5.35×10^{-2} against the Quadrature benchmark, consistent with simulation noise at $N_{sim} = 10^7$.

Moment diagnostic.

The theoretical moments of the Inverse Gaussian mixing variable are validated against the empirical moments of 10^7 Monte Carlo draws (Table 5). All four moments match closely, including the fourth (excess kurtosis, theoretical 14.9590 vs. empirical 14.9448), confirming that the Michael-Jones-Schucany sampler is correctly parametrized. This Monte-Carlo-based check is not exposed to the floating-point cancellation that affects a finite-difference estimate of higher-order moments from the characteristic function.

Moment	Theoretical	Empirical (MC)
Mean	1.000000	0.999877
Variance	0.997268	0.996940
Skewness	2.995899	2.997296
Excess Kurtosis	14.959016	14.944767

Table 5: Inverse Gaussian mixing variable: theoretical vs. Monte Carlo moments ($N_{sim} = 10^7$).

$\alpha = 1/3$ model.

For the facultative $\alpha = 1/3$ case, FFT vs. Quadrature disagree by up to 3.73, more than three orders of magnitude worse than the NIG ($\alpha = 1/2$) case (6.60×10^{-4}); the two FFT parameterizations agree with each other to 1.03×10^{-3} . This is consistent with the slower Fourier-domain decay of the $\alpha = 1/3$ characteristic function, $|\phi(u)| \sim \exp(-c|u|^{2/3})$ versus NIG's $\exp(-c|u|)$: more tail mass survives at high frequency, making the FFT truncation error larger and confirming that the numerical grid parameters must be recalibrated according to the tail decay of the chosen Lévy model.

5. Case study: Volatility Surface Calibration

a. Methodology

A Normal Tempered Stable model ($\alpha = 2/3$) is calibrated to the Eurostoxx 50 implied volatility surface. Model prices are computed by quadrature and calibrated by minimizing a penalized sum of

squared pricing errors with constant weights,

$$\min_{\sigma, \kappa, \eta} \sum_{i=1}^N (C_{\text{model}}(K_i; \sigma, \kappa, \eta) - C_{\text{mkt}}(K_i))^2 + \mathcal{P}(\sigma, \kappa, \eta), \quad (5)$$

with `fminsearch` (unconstrained Nelder-Mead) and a large fixed penalty whenever a candidate violates $\sigma > 0$, $\kappa > 0$, or the model's admissibility bound on η .

Implied volatilities are then backed out from the calibrated model prices via a vectorized inversion of the Black-76 formula. Rather than MATLAB's built-in `blsimpv`, the inversion is solved with `fsolve` in log-volatility space ($y = \log v$, $v = e^y$), which guarantees a strictly positive implied volatility by construction regardless of the solver path, instead of relying on an externally imposed bound.

b. Results and Discussion

Parameter	Symbol	Value
Volatility level	$\hat{\sigma}$	0.1374
Vol-of-vol	$\hat{\kappa}$	0.2564
Skewness	$\hat{\eta}$	25.2061
Objective value	$\hat{f}$	5.8249

Table 6: Calibrated NTS model parameters ($\alpha = 2/3$).

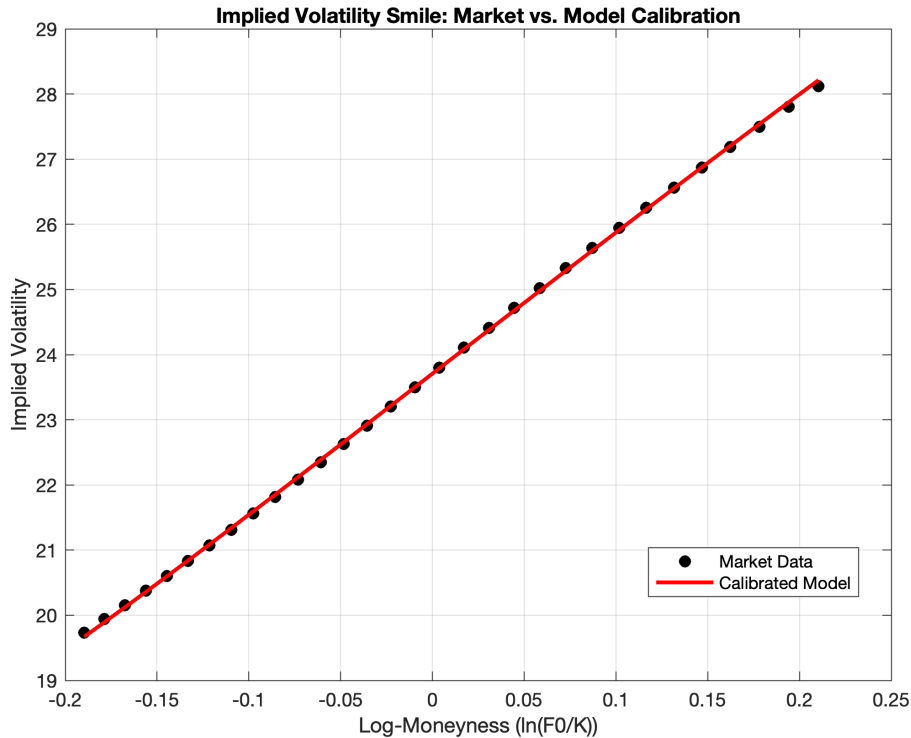


Figure 4: Implied volatility smile: market vs. calibrated NTS model, against log-moneyness $x = \log(F_0/K)$.

The calibrated model tracks the market smile closely across the full moneyness range, with the fitted skewness parameter $\hat{\eta} = 25.21$ producing an implied volatility that rises monotonically as log-moneyness increases – i.e. lower volatility for OTM calls, higher volatility for OTM puts – the same negative skew pattern already observed on the same underlying in Exercise 2 ($\partial\sigma/\partial K < 0$ at the

ATM point). This is consistent with the structural market demand for downside protection on the Eurostoxx 50.